

Erdős Problem #501

EP-501

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

For every $x \in \mathbb{R}$ let $A_x \subset \mathbb{R}$ be a bounded set with outer measure < 1 . Must there exist an infinite independent set, that is, some infinite $X \subseteq \mathbb{R}$ such that $x \notin A_y$ for all $x, y \in X$?

If the sets A_x are closed and have measure < 1 , then must there exist an independent set of size 3?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>